Experiment 5

Looping Constructs

Aim / Objective:

The aim of this experiment is to understand and implement various looping constructs in Java, such as for, while, and do-while loops. Looping is a fundamental concept in programming that allows a set of instructions to be executed repeatedly based on a condition. Through this experiment, students will learn how to use loops to perform repetitive tasks efficiently, which is essential for writing optimized and concise code. The objective is to enable students to identify the appropriate type of loop to use depending on whether the condition is checked before or after the loop body is executed. By practicing different looping constructs, students will gain hands-on experience in writing programs that involve tasks like pattern printing, summation, factorial calculation, and more. This will strengthen their logical thinking and problem-solving abilities in Java programming..

.Problem Statement:

1. **Program to display multiplication table**
2. **Program to check for Armstrong number** **Theory**

/ Concept:

In Java, looping constructs are used to execute a block of code repeatedly until a specified condition is met. Loops help reduce redundancy, improve code efficiency, and simplify repetitive tasks. Java provides three main types of loops: for, while, and do-while, each suited for different situations.

1. **For Loop**

The for loop is an entry-controlled loop used when the number of iterations is known beforehand. It consists of three parts: initialization, condition, and increment/decrement. The syntax is:

```
for(initialization; condition; update) {
   // code to be executed
}
```

2. **While Loop**

The while loop is also an entry-controlled loop used when the number of iterations is not known in advance and depends on a runtime condition. It checks the condition before executing the loop body. The syntax is:

```
while(condition) {
   // code to be executed
}
```

3. Do-While Loop

The do-while loop is an exit-controlled loop, which means the loop body is executed at least once before the condition is tested. It is useful when the loop must run at least one time regardless of the condition. The syntax is:

```
do {
   // code to be executed
} while(condition);
```

Each loop has its use case. The choice of loop depends on the problem's requirements—whether the number of repetitions is known or depends on user input or dynamic conditions. Understanding and mastering these looping constructs is essential for solving a wide variety of programming problems in Java.

Algorithm :

Program to display multiplication table Start

1. Start
2. Declare an integer variable `num`
3. Accept a number from the user and store it in `num`
4. Use a for loop from i = 1 to 10
   a. Inside the loop, calculate num * i
   b. Display the result in the format: `num x i = result`
5. End

Program to check for Armstrong number Start

1. Start
2. Declare integer variables: num, originalNum, remainder, and result
3. Accept a number from the user and store it in num
4. Assign originalNum = num
5. Initialize result = 0
6. Repeat while num != 0:
   a. Get the last digit: remainder = num % 10
   b. Cube the digit and add to result: result += remainder * remainder * remainder

   c. Remove the last digit: num = num / 10
7. After the loop, compare originalNum with result:
   a. If equal, it's an Armstrong number
   b. Else, it's not
8. End

Program code:

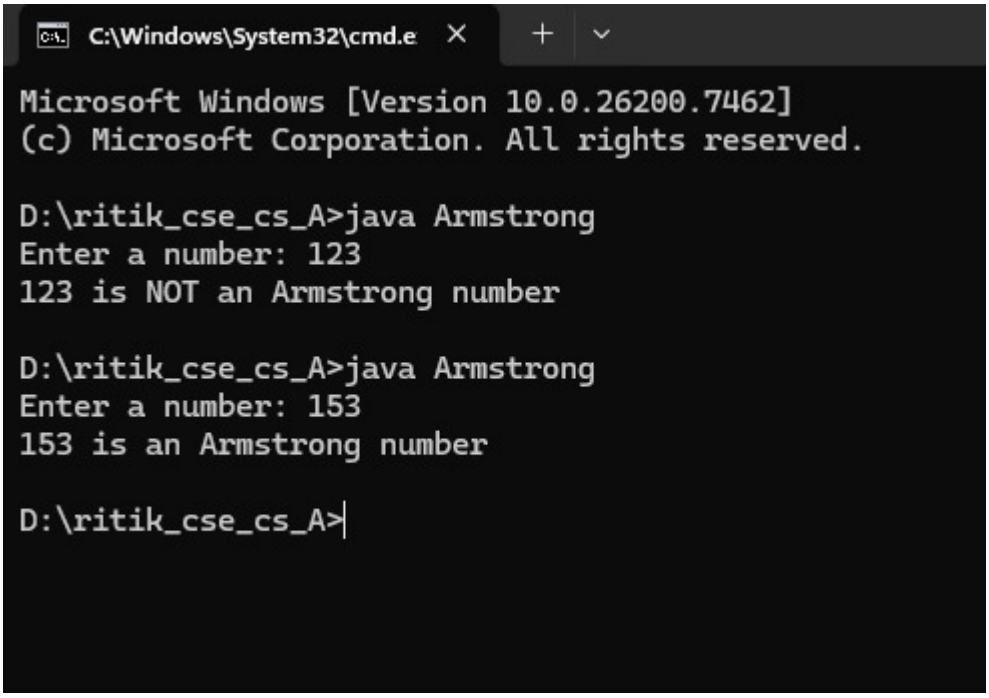
1. Program to display multiplication table

Output:

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

D:\ritik_cse_cs_A>javac MultiplicationTable.java

D:\ritik_cse_cs_A>java MultiplicationTable
Multiplication table of 2:
2x0 = 0
2x1 = 2
2x2 = 4
2x3 = 6
2x4 = 8
2x5 = 10
2x6 = 12
2x7 = 14
2x8 = 16
2x9 = 18
2x10 = 20

D:\ritik_cse_cs_A>D
```

2. Program to check for Armstrong number

Output:



Learning Outcomes:

Students will able to

LO1 Use different looping constructs (for, while, do-while) in Java.
LO2 Implement logic for repetitive tasks like multiplication tables and Armstrong number checks
LO3 Select appropriate loops based on problem requirements.

Course Outcome:

CO3: Apply looping constructs in Java to develop efficient solutions for problems involving repetitive tasks.

Conclusion:

In this experiment, students successfully explored and implemented various looping constructs in Java, including for, while, and do-while loops. By developing programs like multiplication tables and Armstrong number checks, they understood how loops can simplify repetitive tasks and enhance code efficiency. This experiment reinforced their logical thinking and helped them learn to choose the appropriate loop based on the problem's requirements.

Viva Questions:

1. What is the difference between while and do-while loops in Java?
2. When would you prefer using a for loop over a while loop?
3. What is an infinite loop? How can it be avoided?
4. How does the break statement work inside a loop?
5. Can a loop be nested inside another loop? Give an example.

For Faculty Use only:

Correction Parameters	**Formative Assessment [40%]**	**Timely completion of Practical [40%]**	**Attendance / Learning Attitude [20%]**	**Total**
Marks Obtained				